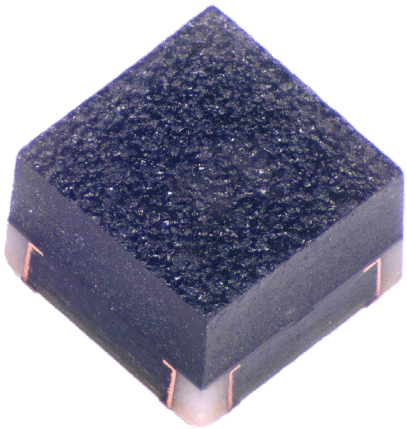
UHD111A-FKA: RGB SMD LED



PRODUCT DESCRIPTION

The UHD111A full-color RGB LED offers a high-intensity light output and a wide viewing angle. The compact 1.0mm x 1.0mm package allows for a very high resolution screen and is designed to work in a wide array of environmental conditions. Cree LED full-color RGB LEDs are suited for indoor video screen, decorative lighting and amusement applications.

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FEATURES

- Size (mm): 1.0 x 1.0
- Dominant Wavelength
  Red (617 - 622nm)
  Green (524 - 528nm)
  Blue (465 - 469nm)
- Luminous Intensity (mcd)
  Red (29-38)@ 5mA
  Green (51-66.5)@ 3mA
  Blue (10.7-14)@ 3mA
- Moisture Sensitivity Level: 5a
- Lead-Free
- RoHS Compliant
- Matte Surface
- High contrast

APPLICATIONS

- Full-Color Video Screen
- Decorative Lighting
- Amusement

Cree LED / 4001 E. Hwy. 54, Suite 2000 / Durham, NC 27713 USA / +1.919.313.5330 / www.cree-led.com

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ABSOLUTE MAXIMUM RATINGS (T_A = 25°C)

Items	Symbol	Absolute Maximum Rating			Unit
		R	G	B	
Forward Current Note 1	I_F	10	10	10	mA
Peak Forward Current Note 2	I_{FP}	30	30	30	mA
Reverse Voltage	V_R	10	10	10	V
Power Dissipation	P_D	25	30	30	mW
Operation Temperature	T_{opr}	-40 ~ +85			°C
Storage Temperature	T_{stg}	-40 ~ +110			°C
Junction Temperature	T_J	115	115	115	°C
Junction/ambient	R_{THJA}	1087	987	841	°C/W
Junction/solder point	R_{THJS}	999	894	735	°C/W

Note:

1. Single-color light
2. Pulse width ≤0.1 msec, duty ≤1/10.

TYPICAL ELECTRICAL & OPTICAL CHARACTERISTICS (T_A = 25°C)

Characteristics	Condition	Symbol	Values			Unit
			R	G	B	
Dominant Wavelength	I_F = 5mA(R) I_F = 3mA(G) I_F = 3mA(B)	λ_{DOM}	617~622	524~528	465~469	nm
Spectral bandwidth at 50% I_{REL} max	I_F = 5mA(R) I_F = 3mA(G) I_F = 3mA(B)	$\Delta\lambda$	24	38	28	nm
Forward Voltage	I_F = 5mA(R) I_F = 3mA(G) I_F = 3mA(B)	$V_{F(min)}$	1.7	2.3	2.3	V
		$V_{F(max)}$	2.3	3.1	3.1	V
Luminous Intensity	I_F = 5mA(R) I_F = 3mA(G) I_F = 3mA(B)	$I_{V(avg)}$	35	60	12	mcd
Reverse Current (max)	V_R = 10 V	I_R	0.5	0.5	0.5	μA

* Tolerance of Luminous Intensity: ±10%.
* Tolerance of Dominant Wavelength: ±1nm.
* Tolerance of Forward Voltage: ±0.1V.
* Continuous reverse voltage can cause LED damage.

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INTENSITY BIN LIMIT

Red (5 mA)			Green (3 mA)			Blue (3 mA)		
Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)
3k2	29	38	3e1	51	66.5	L3	10.7	14

* Tolerance of measurement of luminous intensity is ±10%.

COLOR BIN LIMIT

Red (5 mA)			Green (3 mA)			Blue (3 mA)		
Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)
RV	617	622	g5z	524	528	b3z	465	469

* Tolerance of measurement of dominant wavelength is ±1 nm.

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ORDER CODE TABLE

Kit Number	Color	Luminous Intensity (mcd)				Dominant Wavelength (nm)				Package
		Color Bin	Min. (mcd)	Color Bin	Max. (mcd)	Color Bin	Min. (nm)	Color Bin	Max. (nm)	
UHD111A-FKA-C3k23e1L3Vg5zb3z3	Red	3k2	29	3k2	38	RV	617	RV	622	Reel
	Green	3e1	51	3e1	66.5	g5z	524	g5z	528	Reel
	Blue	L3	10.7	L3	14	b3z	465	b3z	469	Reel

Notes:

- The above kit numbers represent order codes that include multiple intensity-bin and color-bin codes. Only one intensity-bin code and one color-bin code will be shipped on each bulk. Single intensity-bin code and single color-bin codes will not be orderable.
- Please refer to the HB LED Lamp Reliability Test Standards document for reliability test conditions.
- Please refer to the HB LED Lamp Soldering & Handling document for information about how to use this LED product safely.

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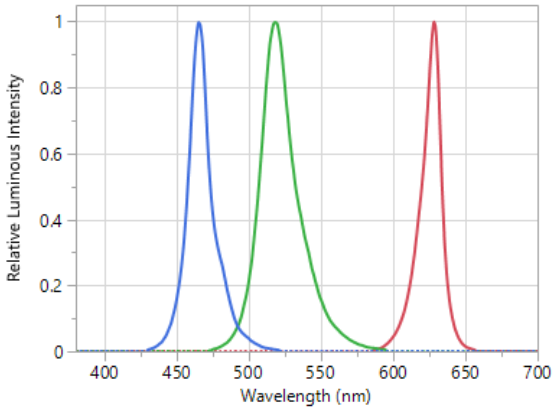
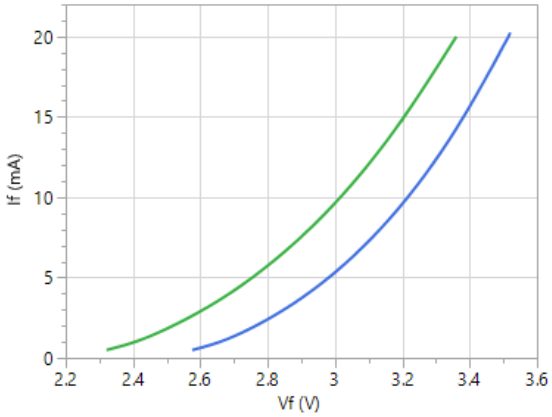
GRAPHS

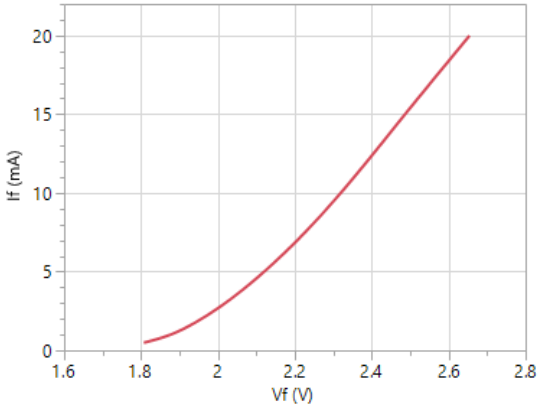
The data below are collected from statistical figures that do not necessarily correspond to the actual parameters of each single LED. Hence, these data will be changed without further notice.



RELATIVE INTENSITY VS. DOMINANT WAVELENGTH



FORWARD CURRENT VS. FORWARD VOLTAGE



FORWARD CURRENT VS. FORWARD VOLTAGE

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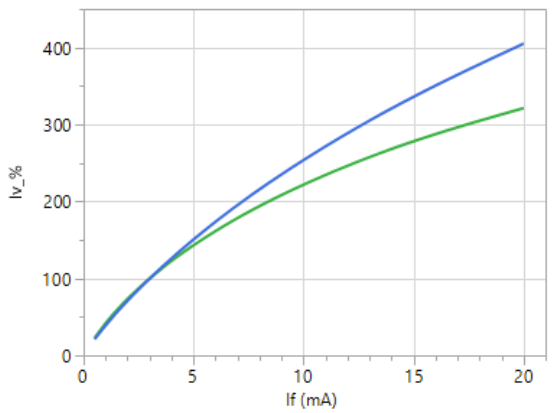
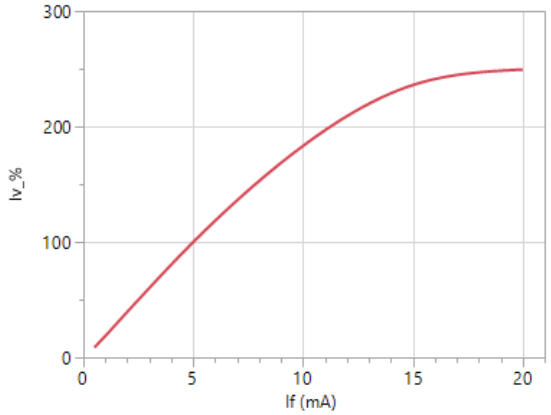
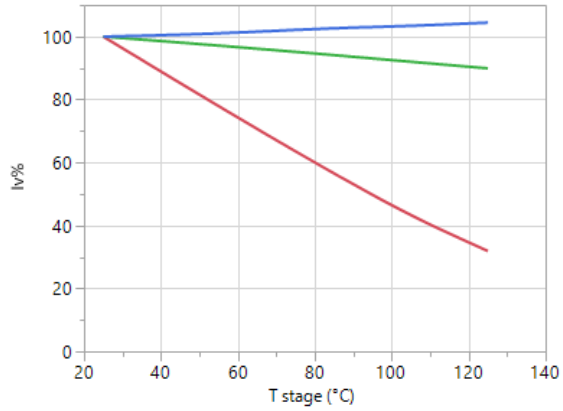


RELATIVE LUMINOUS INTENSITY VS. FORWARD CURRENT



RELATIVE LUMINOUS INTENSITY VS. FORWARD CURRENT



RELATIVE LUMINOUS INTENSITY VS. TEMPERATURE

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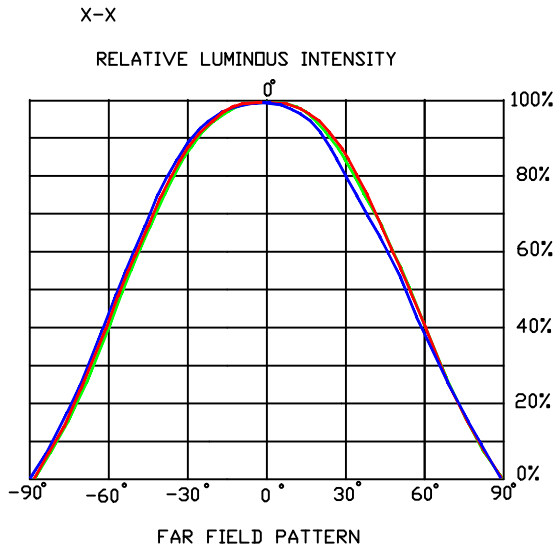

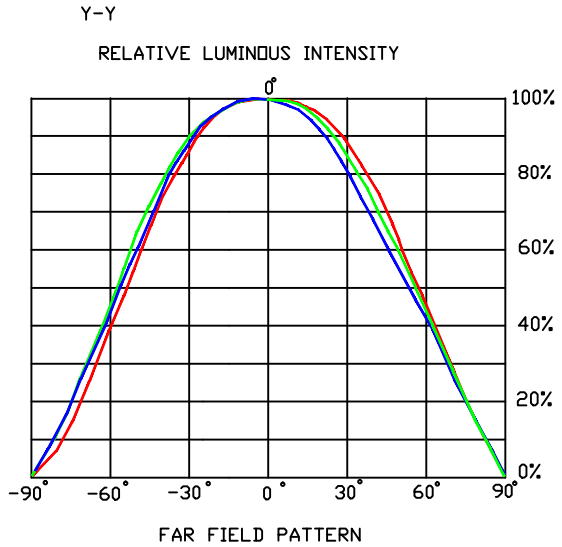

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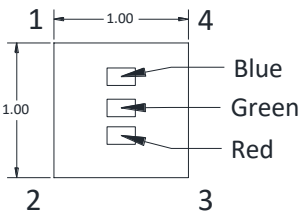
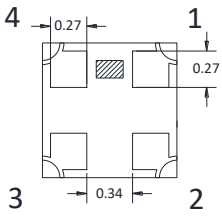
MECHANICAL DIMENSIONS

All dimensions are in mm.

Tolerance of measurement of the dimension is ±0.1.

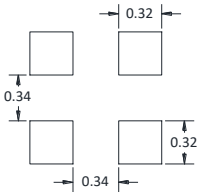


Top View

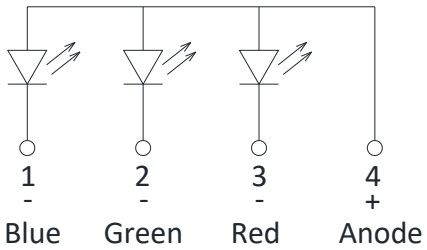


Bottom View

Side View



PCB Footprint



NOTES

RoHS Compliance

The levels of RoHS restricted materials in this product are below the maximum concentration values (also referred to as the threshold limits) permitted for such substances, or are used in an exempted application, in accordance with EU Directive 2011/65/EC (RoHS2), as implemented January 2, 2013. RoHS Declarations for this product can be obtained from your Cree LED representative or from the Product Ecology section of the Cree LED website.

Vision Advisory

WARNING: Do not look at an exposed lamp in operation. Eye injury can result.

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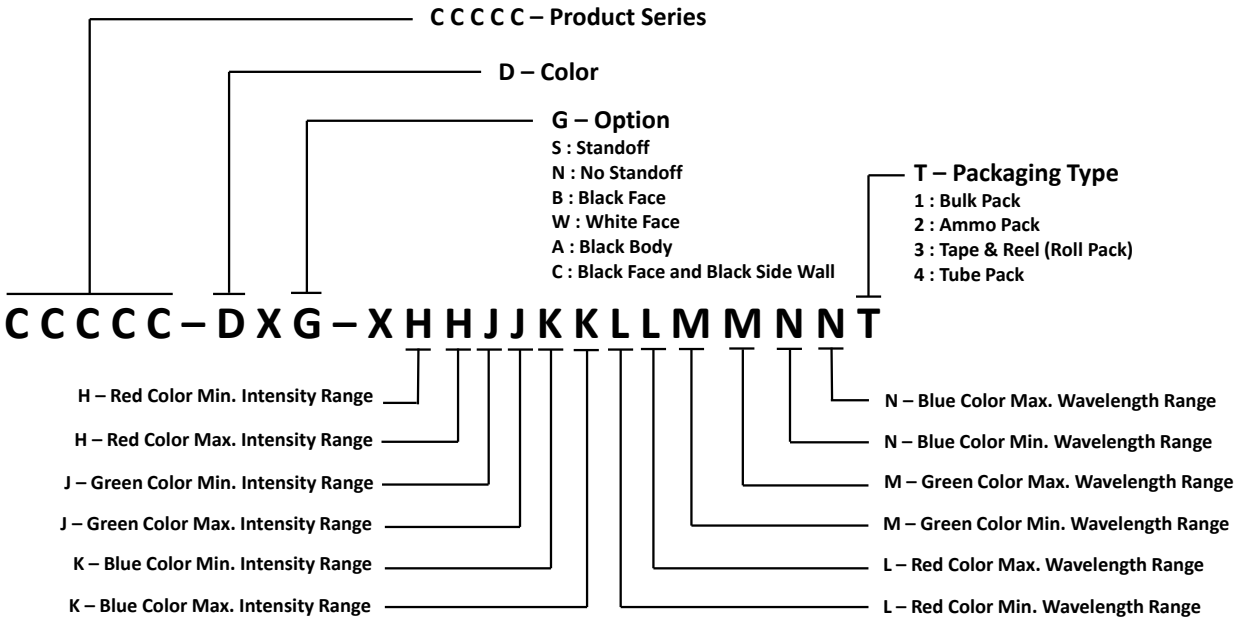
KIT NUMBER SYSTEM

Cree LED lamps are tested and sorted into performance bins. A bin is specified by ranges of color, forward voltage, and brightness.

Cree LEDs are sold by order codes in combinations of bins called kits. Order codes are configured in the following manner:



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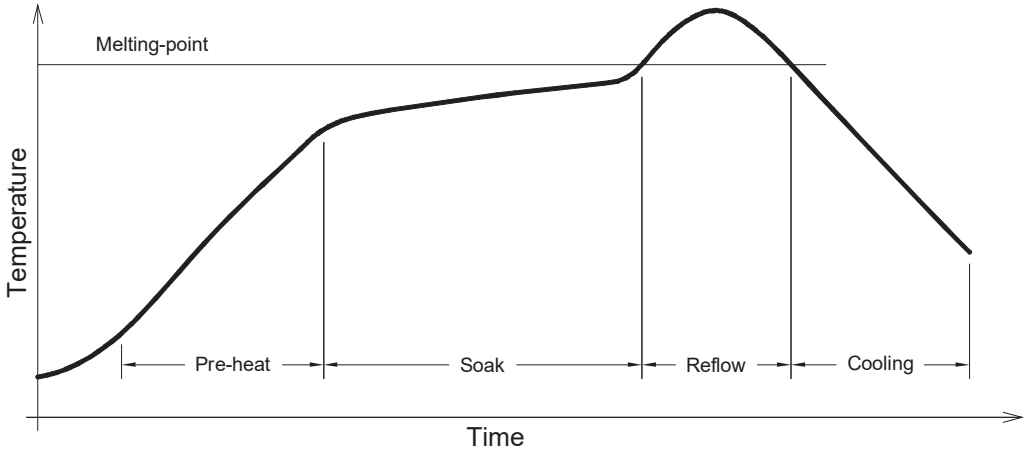
REFLOW SOLDERING

- The UHD111A-FKA is rated as a MSL 5a product.
- After opening the sealed bag,the SMD LED must be stored under the condition<30°C and<60%RH. Under these conditions,the SMD LEDs must be used (subject to reflow) within 24 hours after bag opening, and baking 4 hours/60°C ±5°C is required when exceeding 24 hours.
- Note that baking must only be done once.
- The temperature profile is as below.

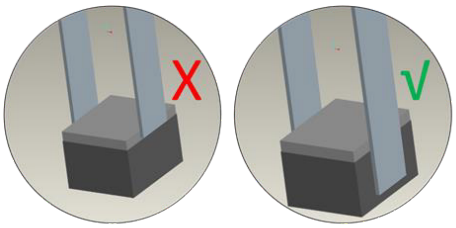


Use only with UHD111A-FKA

Solder
Average ramp-up rate = 4 °C/second max.
Soak temperature = 183-217°C
Soak time = 120 seconds max.
Duration above 217 °C = 60 seconds max.
Peak temperature = 230-240°C max
Time within 5 °C of peak temperature = 10 seconds max.
Ramp-down rate = 6 °C/second max.

NOTES

- The packaging sizes of this model is very small and the resin is still soft after solidification. Users are required to handle with care. Never touch the resin surface of SMD products.
- To avoid damaging the product's surface and interior device, it is recommended to choose a special nozzle to pick up the SMD products during the process of SMT production.touch the package by hand is not suggested and avoid scratch on device surface. The following method is necessary:



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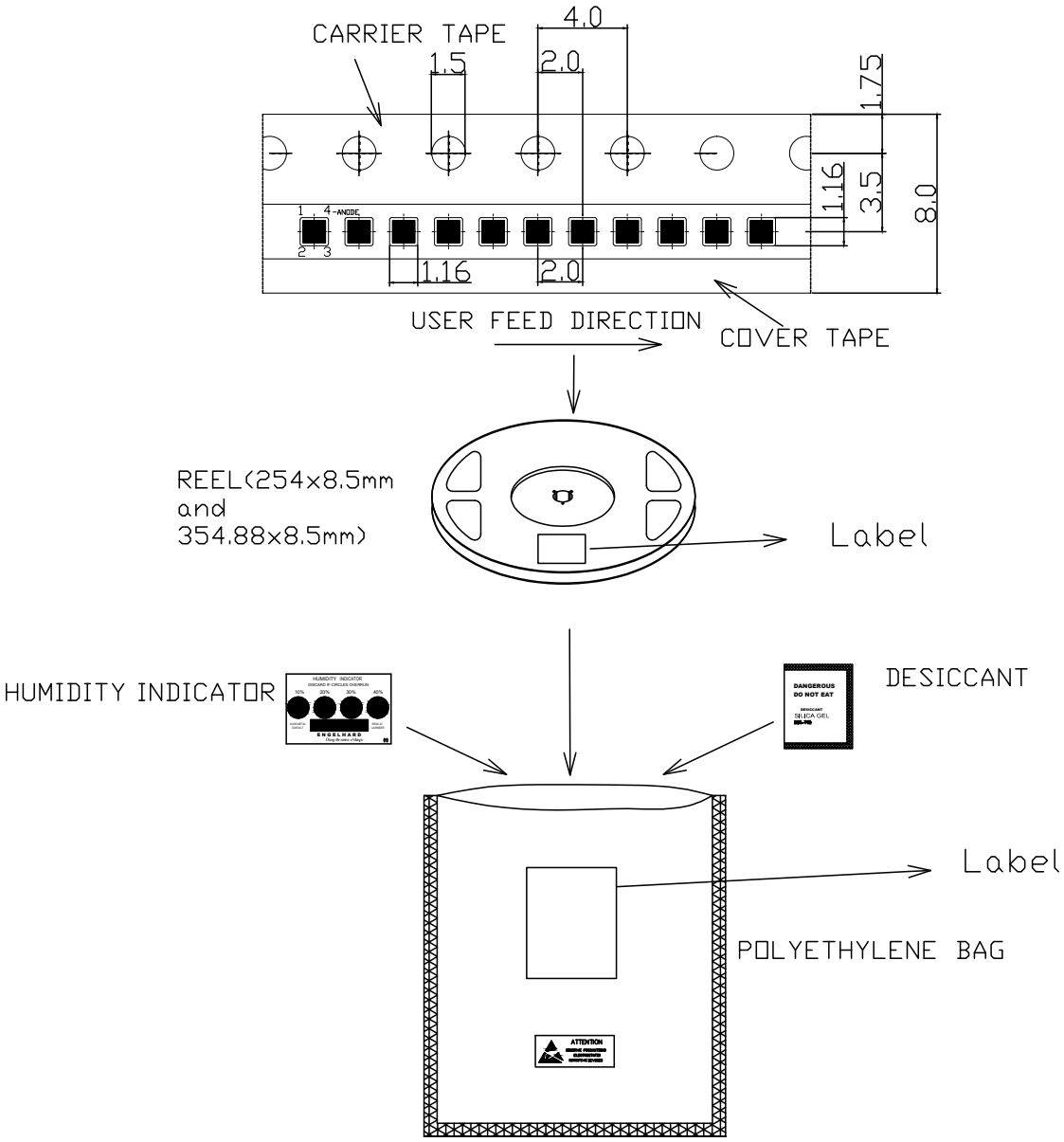
PACKAGING

- The boxes are not water resistant and they must be kept away from water and moisture.
- The LEDs are packed in cardboard boxes after packaging in normal or anti-electrostatic bags.
- Cardboard boxes will be used to protect the LEDs from mechanical shocks during transportation.
- The reel pack is applied in SMD LED.
- Max 42000 pcs per reel.



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